# Anaïs Azouaoui

MSc student in Computer Science, seeking a 6-month research internship from March 2027

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*Research interests: robot learning, imitation and reinforcement learning, vision-language models, representation learning*

## Education

**Master's in Computer Science**, Sorbonne University, Paris — *2025 – 2027 (expected)*
*Algorithms, AI, Interactions & Decision (AI2D)*

**BSc in Computer Science**, Sorbonne University, Paris — *2022 – 2025*

## Research Experience

**Interpretable ML for Intensive Care** | *Research Internship* — Jun – Aug 2026
*IRIT, REVA team - CHU de Toulouse, Toulouse* — Supervisors: G. Nadizar, S. Cussat-Blanc, D. Wilson

- Modeled the severity of 4 respiratory disorders, as rated by clinicians, in 164 mechanically ventilated ICU patients.
- Built a **Cartesian Genetic Programming** engine with **NSGA-II** to evolve short, readable formulas.
- Obtained 2–3-variable formulas with test $R^2$ of 0.62–0.87, on par with the best black-box model (Extra Trees, 0.65–0.83).

**Real-to-Sim Scene Generation** | *Research Project* — Feb – May 2026
*ISIR, Sorbonne University, Paris* — Supervisors: S. Doncieux, E. Garrabe

- Designed a multi-step **LLM/VLM** pipeline converting a text prompt or a photo into a physically valid 3D **Genesis** scene.
- Back-projected **GroundingDINO** boxes with metric **Depth Anything V2**, cutting real-photo 3D error from 1.25 to 0.48 m.
- Fine-tuned **Phi-3-mini** with 4-bit **QLoRA** (Unsloth, TRL), raising spatial-relation F1 from 56% (GPT-4o) to 72%.
- Mapped free-text object names (synonyms, French or English) to 3D assets by **embedding retrieval** with **multilingual-E5**.
- Built 3,531 synthetic text–scene pairs filtered by an **LLM verifier**, and a **VLM loop** correcting the rendered scene.

## Projects

**Damage Recovery for a Quadruped Robot** — 2026 | Project page

- Reproduced **Intelligent Trial & Error** (ITE) for few-trial damage recovery on a simulated **Brax** quadruped.
- Built gait repertoires with evolution (**MAP-Elites**) and evolution plus RL (**PGA-ME**, **DCRL-ME**) in **QDax** (**JAX**).
- Recovered in a median of 5 trials across 12 types of leg damage with **Bayesian optimization**, nearly doubling speed.
- Measured 90× fewer episodes for ITE than for **TD3** fine-tuning to adapt to the same damages.

**RoboSimon: Memory in Visuomotor Imitation Policies** — 2026 | Project page

- Built a Simon game in **MuJoCo** where the robot sees a sequence of 2 to 4 lit buttons and must repeat it from memory.
- Trained **ACT** and **Diffusion Policy** with a shared **transformer** memory, so that only the action head differs.
- Found ACT reaches 100% while Diffusion Policy needs 3× more training to go from 34–78% to 99.7%.
- Showed with **counterfactual tests** and **linear probes** that its memory failures came from under-training.

**Vision-Language-Action Model Robustness** — 2025 | Project page

- Evaluated **SmolVLA** on 3,688 **LIBERO** episodes under visual and arm perturbations: a 3° joint offset halves success.
- Paraphrased instructions (**metamorphic testing**) cut success from 80% to 23%; fixed by **sentence-encoder** retrieval.
- Searched with **MAP-Elites** for mixes of small perturbations, harmless alone, that fail 71% of held-out episodes.

**Camera-Guided Navigation of a Self-Built Wheeled Robot** — 2026 | Project page

- Built a **C++17** navigation stack with **EKF** state estimation, A* motion planning and **MPPI** model predictive control.
- Ran the same code in simulation and on hardware, fixing the **sim-to-real** gap (static friction, brownouts, camera latency).
- Reached 10/10 real goals at 2 cm; MPPI solved 143 of 145 simulated goals without collision, against 116 for Pure Pursuit.

## Skills

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| **Domains** | Deep Learning, Reinforcement Learning, Computer Vision, LLMs/VLMs, NLP, Robot Learning, Motion Planning |
| **Methods** | Transformers, QLoRA, Embedding Retrieval, Imitation Learning, Evolutionary Computation, EKF, MPPI |
| **Tools** | Python, C++17, PyTorch, JAX, Hugging Face, OpenCV, scikit-learn, Simulation (Genesis, LeRobot), Git, Docker |
| **Languages** | French (native), English (proficient), Arabic (proficient), Japanese (basic) |

## Publications

[1] **A. Azouaoui**, C. Zouloumian, E. Garrabe, S. Doncieux. *Real-to-Sim: generating physically valid robot simulation scenes from text and images with LLMs and VLMs*. In preparation.

[2] **A. Azouaoui**, G. Nadizar, S. Cussat-Blanc, D. Wilson, M. Poette. *Interpretable prediction of clinician-rated respiratory severity in intensive care with genetic programming*. In preparation.